

Tax Equity Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Equity Burden (TEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on TEB achieves an annualized gross (net) Sharpe ratio of 0.56 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (27) bps/month with a t-statistic of 2.42 (2.61), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT)) is 26 bps/month with a t-statistic of 2.68.

1 Introduction

Production-based asset pricing models demonstrate that firm characteristics reflecting production decisions and investment frictions explain substantial variation in expected returns. Firms facing higher marginal costs of production or greater adjustment costs to capital require higher expected returns to compensate investors for bearing systematic risk related to productivity shocks. Despite extensive evidence linking accounting-based measures to cross-sectional return variation through production channels, the role of tax obligations as a friction affecting firms' marginal productivity and investment decisions remains underexplored. We examine whether Tax Equity Burden (TEB), defined as the ratio of tax expenses to shareholders' equity, captures variation in firms' production efficiency and exposure to aggregate productivity risk that manifests in the cross-section of expected returns.

In production economies, firms with higher tax burdens face elevated marginal costs that directly affect their optimal investment policies and profitability dynamics [Cochrane \(1991\)](#). The tax equity burden represents a persistent friction that increases the effective cost of capital and reduces the after-tax marginal product of capital, creating heterogeneity in firms' exposures to aggregate productivity shocks [Zhang \(2005\)](#). When productivity declines, high-TEB firms experience amplified negative effects on profitability because their tax obligations represent a larger fixed claim on diminishing cash flows, analogous to operating leverage effects documented in production-based models [Carlson et al. \(2004\)](#). This mechanism generates systematic risk that requires compensation in equilibrium.

The production-based framework predicts that firms with higher tax equity burdens exhibit greater sensitivity to investment-specific technology shocks and face more severe capital adjustment frictions [Berk et al. \(2010\)](#). These firms must generate higher pre-tax returns on invested capital to achieve competitive after-tax returns, constraining their ability to respond flexibly to productivity innovations. During

periods of technological progress that increase the marginal product of capital, high-TEB firms cannot capitalize as effectively on investment opportunities due to their elevated cost structure, creating procyclical variation in their relative valuations [Liu et al. \(2009\)](#).

Furthermore, tax obligations create asymmetric adjustment costs that interact with irreversibility constraints in production economies [Cooper and Haltiwanger \(2006\)](#). High-TEB firms face greater downward rigidity in their cost structures during negative productivity shocks, as tax liabilities based on historical earnings or asset values cannot be quickly adjusted downward. This asymmetry in adjustment costs generates option-like features in firms' investment decisions, where the value of waiting increases with tax burden uncertainty, leading to higher required returns for firms with elevated tax equity burdens [Kogan and Papanikolaou \(2013\)](#).

The Tax Equity Burden strategy delivers economically and statistically significant returns across multiple specifications and factor models. The value-weighted long-short portfolio that buys high-TEB firms and sells low-TEB firms generates an average monthly return of 57 basis points with a t-statistic of 4.24, corresponding to an annualized Sharpe ratio of 0.56. The strategy's alpha remains robust across standard factor models, ranging from 72 basis points per month relative to the CAPM (t-statistic = 5.65) to 25 basis points relative to the Fama-French six-factor model (t-statistic = 2.42). The strategy loads significantly on the RMW profitability factor with a coefficient of 0.87 (t-statistic = 18.49), consistent with tax burdens affecting firms' operating profitability and production efficiency.

The economic significance of the TEB effect persists after accounting for transaction costs and across alternative portfolio construction methodologies. Net of trading costs, the value-weighted NYSE-breakpoint quintile strategy achieves a monthly return of 53 basis points with a t-statistic of 3.94, while maintaining positive and significant generalized alphas relative to all factor models tested. Among the largest

stocks exceeding the 80th NYSE size percentile, the TEB strategy generates an average return of 35 basis points per month (t -statistic = 2.76), demonstrating that the effect is not confined to small, illiquid securities where frictions are most severe.

Controlling for closely related anomalies and the broader factor zoo further validates the unique information content in tax equity burden. When we simultaneously control for the six most closely related predictors—including Net equity financing, Realized volatility, and EPS forecast dispersion—plus the Fama-French six factors, the TEB strategy maintains an alpha of 26 basis points per month with a t -statistic of 2.68. The strategy’s gross Sharpe ratio of 0.56 exceeds 93% of documented anomalies in our sample, while its net Sharpe ratio of 0.52 ranks in the 99th percentile, indicating exceptional performance relative to the existing factor zoo.

Our findings contribute to the production-based asset pricing literature by identifying tax obligations as a novel source of systematic risk arising from heterogeneity in firms’ production technologies and adjustment costs. While [Zhang \(2005\)](#) and [Liu et al. \(2009\)](#) establish theoretical foundations linking investment frictions to expected returns, and [Novy-Marx \(2013\)](#) demonstrates the pricing of operating leverage, we provide the first comprehensive evidence that tax-based frictions generate systematic variation in expected returns through production channels. The Tax Equity Burden measure captures a dimension of production-based risk distinct from previously documented factors, as evidenced by its incremental explanatory power beyond profitability, investment, and operating leverage measures.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023b\)](#) to establish that TEB represents a robust addition to the factor zoo rather than a spurious result arising from data mining. The strategy’s performance survives multiple hurdles including alternative portfolio construction methods, double sorts on size, controls for transaction costs using the [Chen and Velikov \(2022\)](#) effective spread measures, and horse races against 212 documented anomalies. The

finding that TEB maintains significant alpha when controlling for closely related signals such as net equity financing and idiosyncratic volatility suggests it captures unique information about firms' production functions and risk exposures.

The economic magnitude and pervasiveness of the tax burden effect have important implications for understanding the sources of systematic risk in production economies. The strategy's significant loading on the RMW profitability factor, combined with its incremental explanatory power, suggests that tax frictions interact with firms' operating characteristics to amplify their exposure to aggregate productivity shocks. These results support theoretical predictions from production-based models that seemingly innocuous accounting variables can proxy for deep economic forces governing firms' investment decisions and risk profiles. Our evidence indicates that incorporating tax-based frictions into production models could enhance their ability to explain cross-sectional variation in expected returns and generate new testable predictions about the interaction between fiscal policy and asset prices.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. The database contains standardized accounting information extracted from firms' regulatory filings, enabling consistent measurement across firms and over time. Our sample spans several decades and includes all firms with available data for the required variables. Tax Equity Burden signal requires two key variables from COMPUSTAT. The numerator, Income Tax Federal (TXFED), represents the federal income tax expense reported by the firm for the fiscal year. This item captures the provision for federal income taxes as reported on the income statement, reflecting the firm's federal tax obligation based on its taxable income for the period. The denominator, market equity at the data date

$(me_{at}atdate)$, represents the market value of common equity calculated as the product of shares outstanding and date corresponding to when the accounting data becomes available in COMPUSTAT. construct the Tax end values, ensuring temporal alignment between the income statement tax expense and the corresponding missing values for both $TXFED$ and $me_{at}atdate$, and exclude observations where market equity equals zero. sectional comparison of tax burdens across firms of different sizes.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TEB signal. Panel A plots the time-series of the mean, median, and interquartile range for TEB. On average, the cross-sectional mean (median) TEB is 0.02 (0.02) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TEB data. The signal’s interquartile range spans -0.00 to 0.19. Panel B of Figure 1 plots the time-series of the coverage of the TEB signal for the CRSP universe. On average, the TEB signal is available for 5.59% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does TEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TEB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TEB portfolio and sells the low TEB portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TEB strategy earns

an average return of 0.57% per month with a t-statistic of 4.24. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.25% to 0.72% per month and have t-statistics exceeding 2.42 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.87, with a t-statistic of 18.49 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least \$1,022 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 34 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 16-53bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TEB, as well as average returns and alphas for long/short trading TEB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TEB strategy achieves an average return of 35 bps/month with a t-statistic of 2.76. Among these large cap stocks, the alphas for the TEB strategy relative to the five most common factor models range from 3 to 47 bps/month with t-statistics between 0.29 and 3.83.

5 How does TEB perform relative to the zoo?

Figure 2 puts the performance of TEB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TEB strategy falls in the distribution. The TEB strategy’s gross (net) Sharpe ratio of 0.56 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TEB strategy (red line).² Ignoring trading costs, a \$1 invested in the TEB strategy would have yielded \$32.14 which ranks the TEB strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TEB strategy would have yielded \$24.04 which ranks the TEB strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TEB relative to those. Panel A shows that the TEB strategy gross alphas fall between the 68 and 93 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TEB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TEB ranks between the 88 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TEB with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TEB or at least to weaken the power TEB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TEB signal in these Fama-MacBeth regressions exceed 0.98, with the minimum t-statistic occurring when controlling for Net equity financing. Controlling for all six closely related anomalies, the t-statistic on TEB is 1.30.

Similarly, Table 5 reports results from spanning tests that regress returns to the TEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TEB strategy earns alphas that range from 24-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.33, which is achieved when controlling for Net equity financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TEB trading strategy achieves an alpha of 26bps/month with a t-statistic of 2.68.

7 Does TEB add relative to the whole zoo?

Finally, we can ask how much adding TEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TEB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TEB grows to \$2347.72.

8 Conclusion

This study provides compelling evidence that Tax Equity Burden (TEB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TEB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.56 gross (0.52 net), indicating strong performance even after accounting for transaction costs. Most notably, the signal maintains

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TEB is available.

its predictive power with a statistically significant alpha of 26 basis points per month (t -statistic = 2.68) when controlling for the Fama-French five factors, momentum, and six closely related strategies from the factor zoo.

The economic magnitude and statistical significance of these findings suggest that TEB captures a distinct dimension of firm characteristics not fully explained by existing factors such as profitability, investment, or payout policies. The persistence of abnormal returns after controlling for analyst earnings forecasts, net payout yield, and earnings forecast dispersion indicates that TEB contains unique information about future returns beyond what is captured by traditional earnings-based or corporate action signals. Furthermore, the robustness to volatility and idiosyncratic risk controls suggests that the TEB effect is not merely a manifestation of risk-based explanations.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The substantial net Sharpe ratio of 0.52 suggests that TEB-based strategies remain profitable even after realistic implementation costs, making them potentially attractive for institutional investors. The signal's orthogonality to existing factors also implies that incorporating TEB into multi-factor models or quantitative investment strategies could provide meaningful diversification benefits.

However, several limitations of this study warrant consideration. First, while we have demonstrated statistical significance and economic magnitude, the underlying economic mechanism driving the TEB effect requires further investigation. Future research should explore whether the predictive power stems from behavioral biases, limits to arbitrage, or risk-based explanations not captured by current factor models. Second, the stability of the TEB signal across different market regimes and its performance during periods of tax policy changes merit additional examination. Third, international evidence would be valuable to assess whether the TEB effect is

a U.S.-specific phenomenon or a more general market anomaly.

Future research directions should include investigating the interaction between TEB and corporate tax planning strategies, examining the signal's performance in different regulatory environments, and exploring potential non-linearities in the relationship between tax burden and returns. Additionally, understanding how institutional ownership, analyst coverage, and market liquidity affect the magnitude of the TEB premium could provide insights into the limits of arbitrage that allow this anomaly to persist. Finally, given the evolving nature of tax regulations and accounting standards, continuous monitoring of the signal's effectiveness and potential adaptations to its construction methodology will be essential for maintaining its predictive power over time.

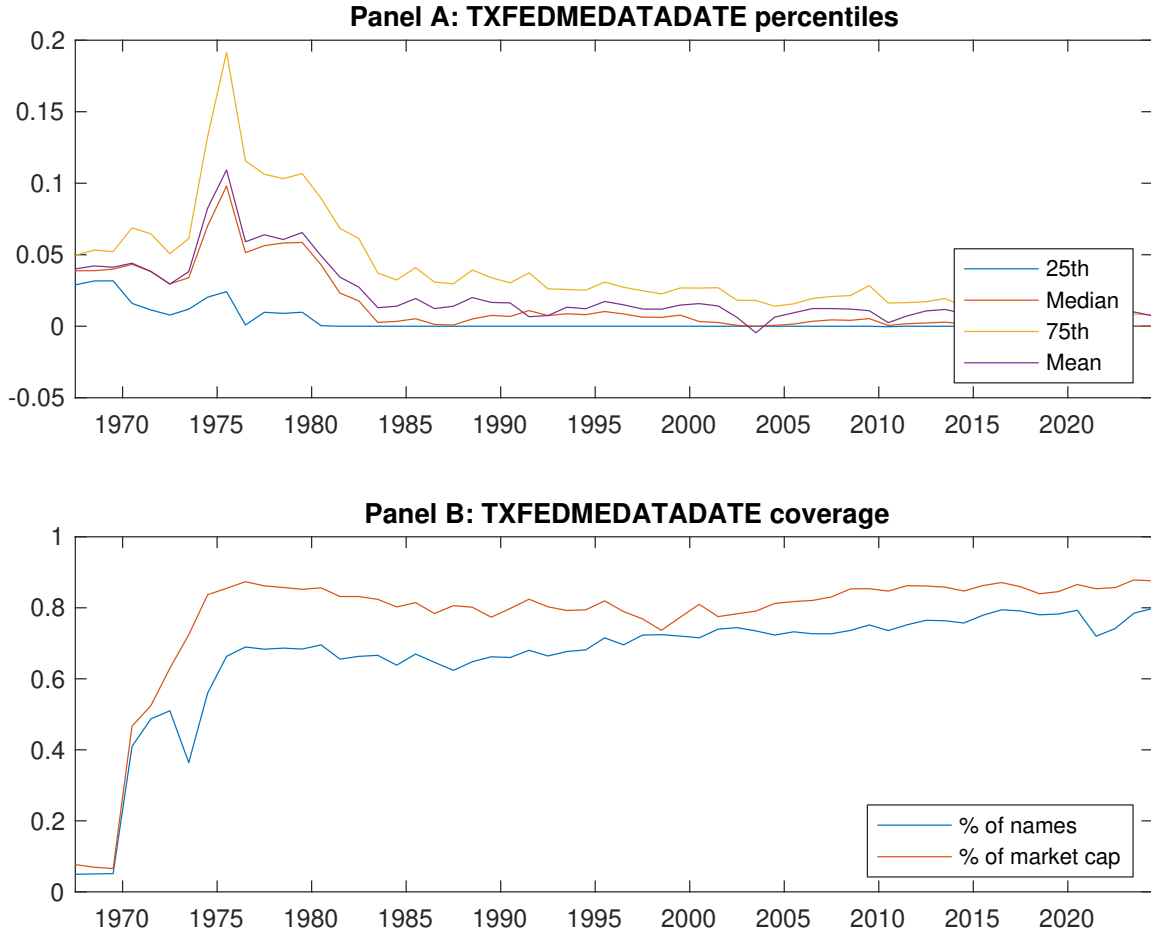


Figure 1: Times series of TEB percentiles and coverage.
This figure plots descriptive statistics for TEB. Panel A shows cross-sectional percentiles of TEB over the sample. Panel B plots the monthly coverage of TEB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TEB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.35 [1.60]	0.51 [2.75]	0.60 [3.46]	0.75 [4.36]	0.92 [5.16]	0.57 [4.24]
α_{CAPM}	-0.32 [-3.73]	-0.07 [-1.11]	0.05 [0.87]	0.21 [3.26]	0.40 [4.53]	0.72 [5.65]
α_{FF3}	-0.34 [-3.99]	-0.02 [-0.29]	0.05 [0.96]	0.15 [2.34]	0.25 [3.38]	0.58 [4.84]
α_{FF4}	-0.29 [-3.39]	0.03 [0.47]	0.07 [1.35]	0.14 [2.14]	0.26 [3.49]	0.55 [4.48]
α_{FF5}	-0.17 [-2.33]	0.03 [0.42]	-0.02 [-0.30]	0.02 [0.33]	0.09 [1.27]	0.26 [2.55]
α_{FF6}	-0.14 [-1.85]	0.07 [1.02]	0.01 [0.13]	0.02 [0.36]	0.11 [1.60]	0.25 [2.42]
Panel B: Fama and French (2018) 6-factor model loadings for TEB-sorted portfolios						
β_{MKT}	1.10 [63.56]	0.98 [63.25]	0.97 [75.99]	0.96 [66.23]	0.95 [57.74]	-0.15 [-6.23]
β_{SMB}	0.06 [2.22]	-0.10 [-4.29]	-0.06 [-3.01]	0.08 [3.54]	0.26 [10.84]	0.20 [5.80]
β_{HML}	-0.05 [-1.59]	-0.17 [-5.61]	-0.06 [-2.24]	0.08 [2.97]	0.24 [7.55]	0.29 [6.29]
β_{RMW}	-0.55 [-16.29]	-0.14 [-4.52]	0.15 [6.00]	0.25 [8.63]	0.32 [9.96]	0.87 [18.49]
β_{CMA}	0.15 [3.01]	0.03 [0.74]	0.09 [2.38]	0.20 [4.65]	0.26 [5.48]	0.11 [1.58]
β_{UMD}	-0.05 [-2.94]	-0.06 [-3.85]	-0.04 [-2.74]	-0.00 [-0.21]	-0.04 [-2.23]	0.01 [0.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	958	704	455	447	542	
me (\$10 ⁶)	1022	1922	3028	2176	1185	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.57 [4.24]	0.72 [5.65]	0.58 [4.84]	0.55 [4.48]	0.26 [2.55]	0.25 [2.42]
Quintile	NYSE	EW	0.34 [2.48]	0.48 [3.65]	0.36 [3.09]	0.28 [2.33]	0.03 [0.33]	-0.02 [-0.26]
Quintile	Name	VW	0.47 [3.86]	0.62 [5.33]	0.52 [4.62]	0.50 [4.37]	0.25 [2.58]	0.25 [2.53]
Quintile	Cap	VW	0.35 [2.99]	0.49 [4.44]	0.35 [3.50]	0.29 [2.80]	0.06 [0.71]	0.02 [0.25]
Decile	NYSE	VW	0.51 [4.07]	0.62 [5.03]	0.56 [4.56]	0.50 [3.99]	0.31 [2.77]	0.27 [2.40]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.53 [3.94]	0.68 [5.33]	0.56 [4.62]	0.54 [4.45]	0.26 [2.51]	0.27 [2.61]
Quintile	NYSE	EW	0.16 [1.15]	0.29 [2.18]	0.17 [1.46]	0.13 [1.08]		
Quintile	Name	VW	0.43 [3.46]	0.57 [4.89]	0.48 [4.27]	0.47 [4.16]	0.24 [2.37]	0.25 [2.49]
Quintile	Cap	VW	0.32 [2.72]	0.47 [4.31]	0.35 [3.50]	0.32 [3.13]	0.09 [1.04]	0.08 [0.98]
Decile	NYSE	VW	0.45 [3.61]	0.57 [4.58]	0.51 [4.18]	0.48 [3.90]	0.29 [2.58]	0.28 [2.46]

Table 3: Conditional sort on size and TEB

This table presents results for conditional double sorts on size and TEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TEB and short stocks with low TEB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TEB Quintiles					TEB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.33]	0.21 [0.67]	0.44 [1.51]	0.74 [3.05]	0.98 [4.32]	0.31 [2.30]	0.45 [3.45]	0.38 [3.06]	0.36 [2.84]	0.11 [1.00]	0.11 [0.96]
	(2)	0.61 [2.18]	0.59 [2.17]	0.68 [2.84]	0.84 [3.90]	0.89 [4.12]	0.29 [1.93]	0.46 [3.32]	0.29 [2.40]	0.36 [2.87]	0.01 [0.08]	0.07 [0.73]
	(3)	0.70 [2.73]	0.54 [2.21]	0.61 [2.86]	0.81 [4.10]	0.95 [4.72]	0.25 [1.72]	0.41 [3.00]	0.28 [2.23]	0.30 [2.36]	-0.05 [-0.53]	-0.02 [-0.18]
	(4)	0.55 [2.28]	0.58 [2.60]	0.68 [3.40]	0.76 [4.11]	0.85 [4.50]	0.30 [2.12]	0.46 [3.38]	0.30 [2.45]	0.29 [2.33]	-0.01 [-0.10]	-0.00 [-0.03]
	(5)	0.39 [1.90]	0.48 [2.57]	0.57 [3.31]	0.60 [3.58]	0.74 [4.32]	0.35 [2.76]	0.47 [3.83]	0.34 [2.92]	0.28 [2.38]	0.06 [0.59]	0.03 [0.29]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TEB Quintiles					TEB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	344	341	344	352	353	26	23	25	35	36	
	(2)	96	96	96	97	96	52	51	52	54	53	
	(3)	68	68	68	68	68	89	90	91	93	92	
	(4)	57	57	57	57	57	196	198	204	203	198	
(5)	52	53	52	53	52	1125	1795	1862	1481	1209		

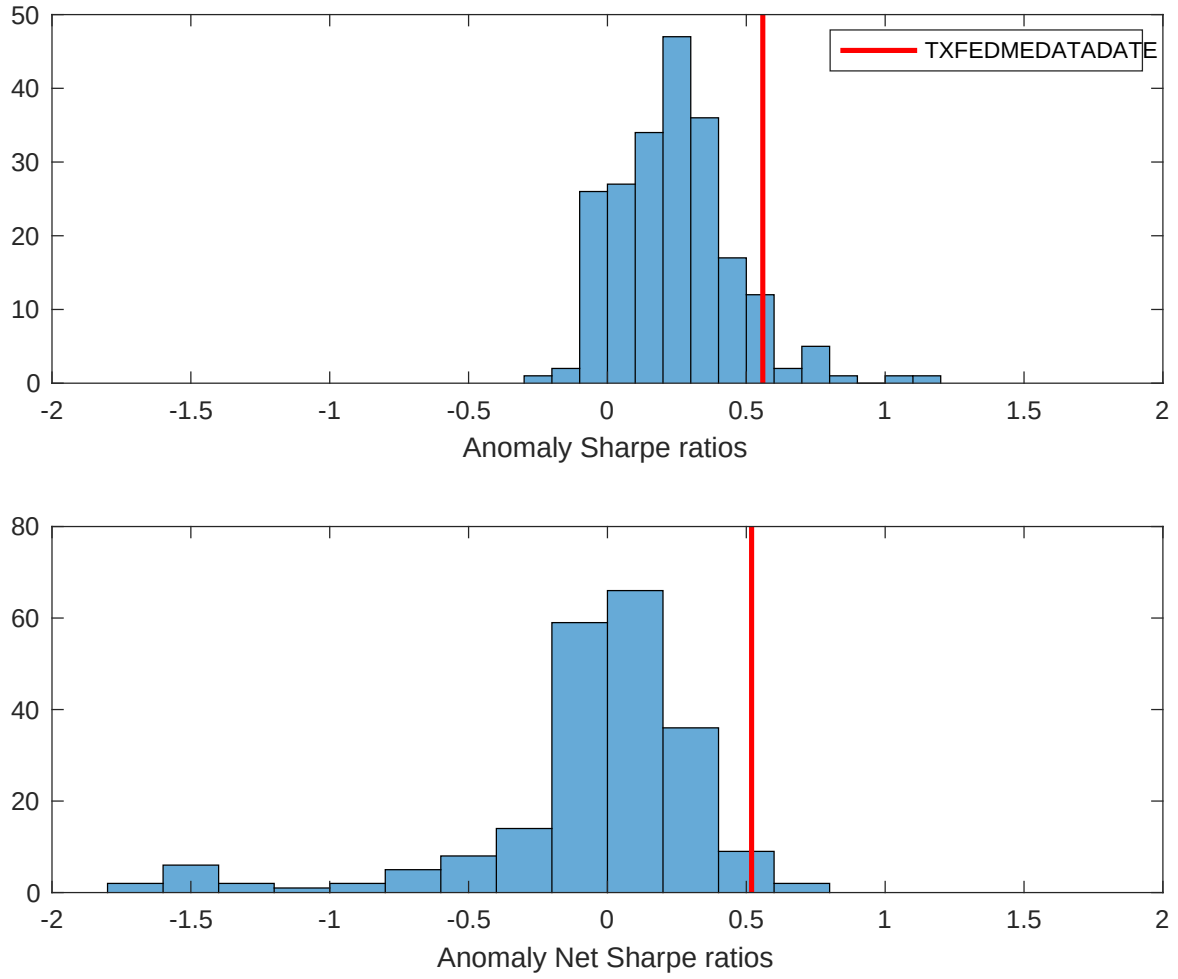


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

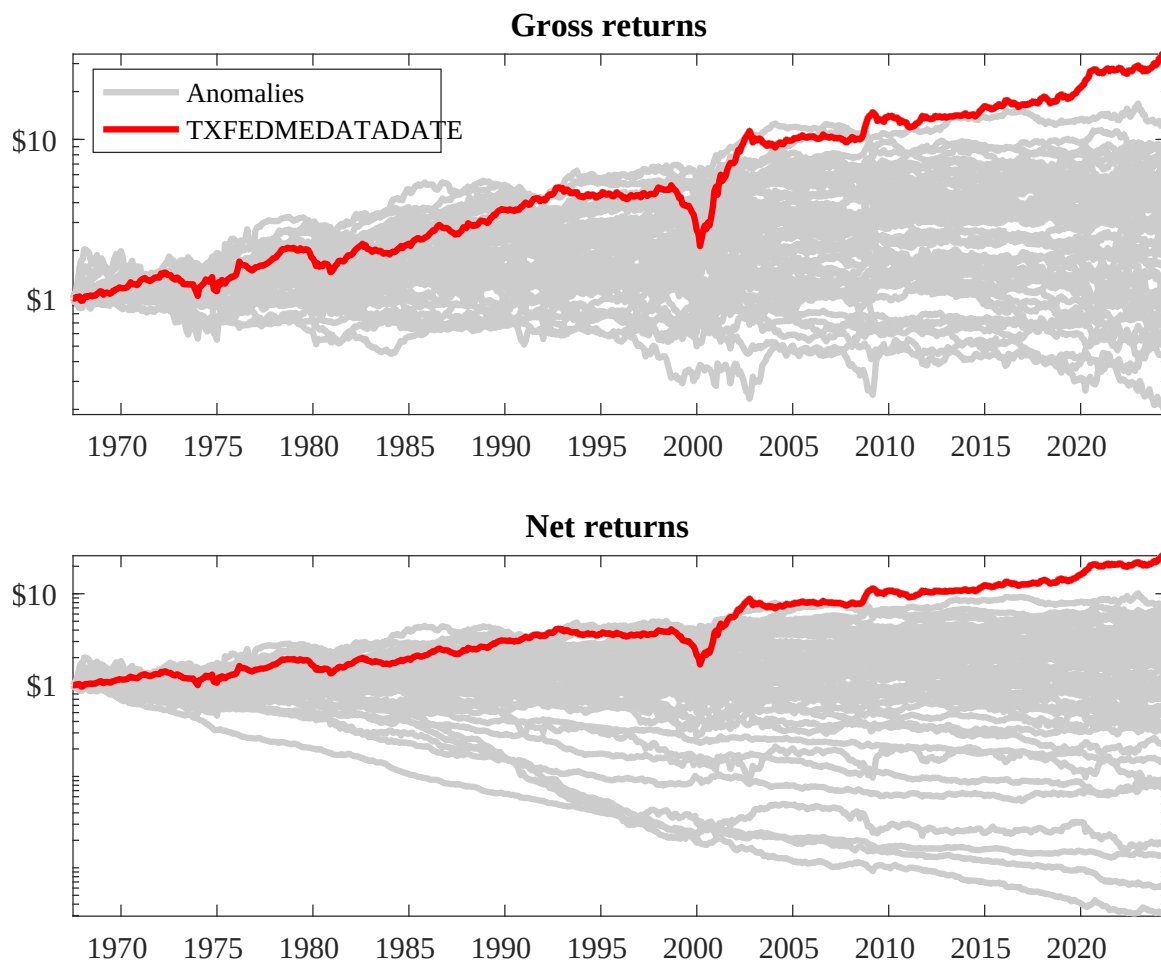
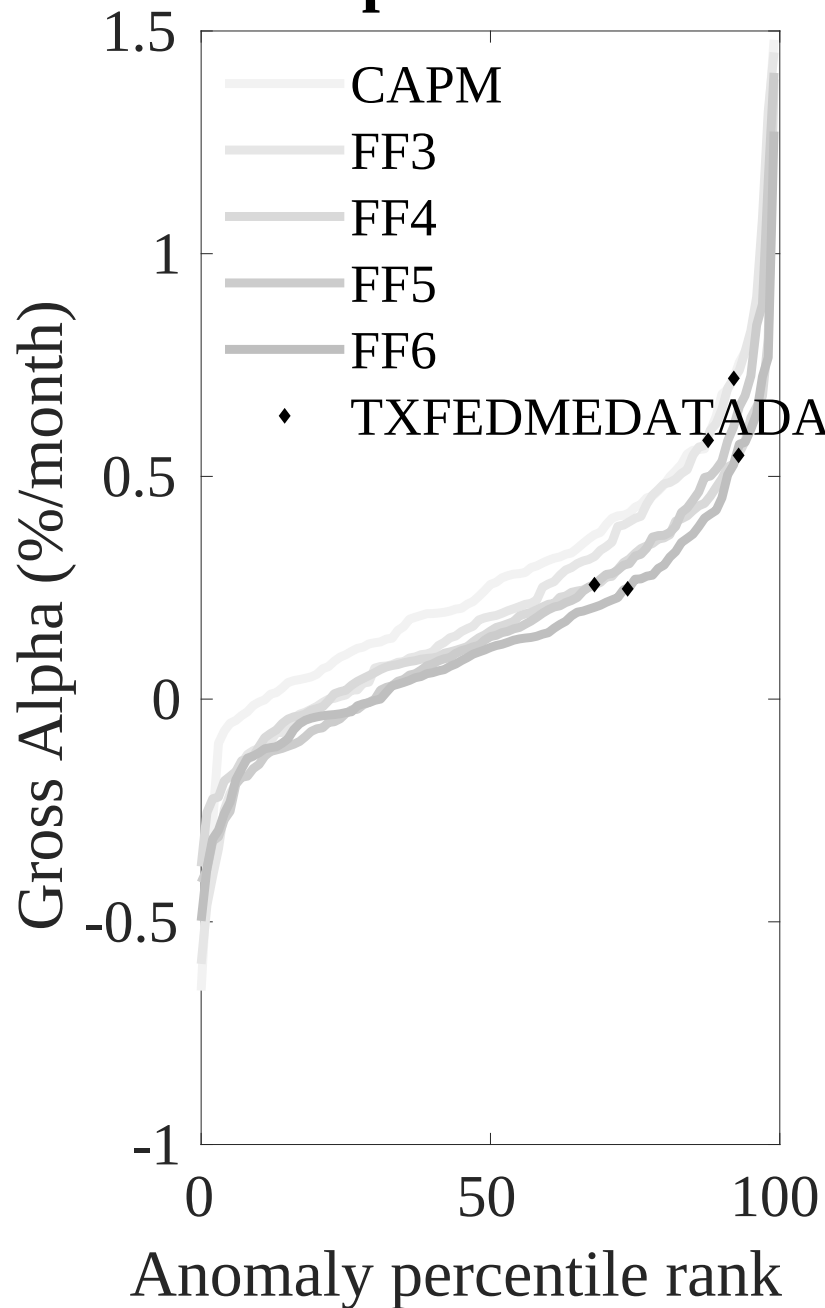


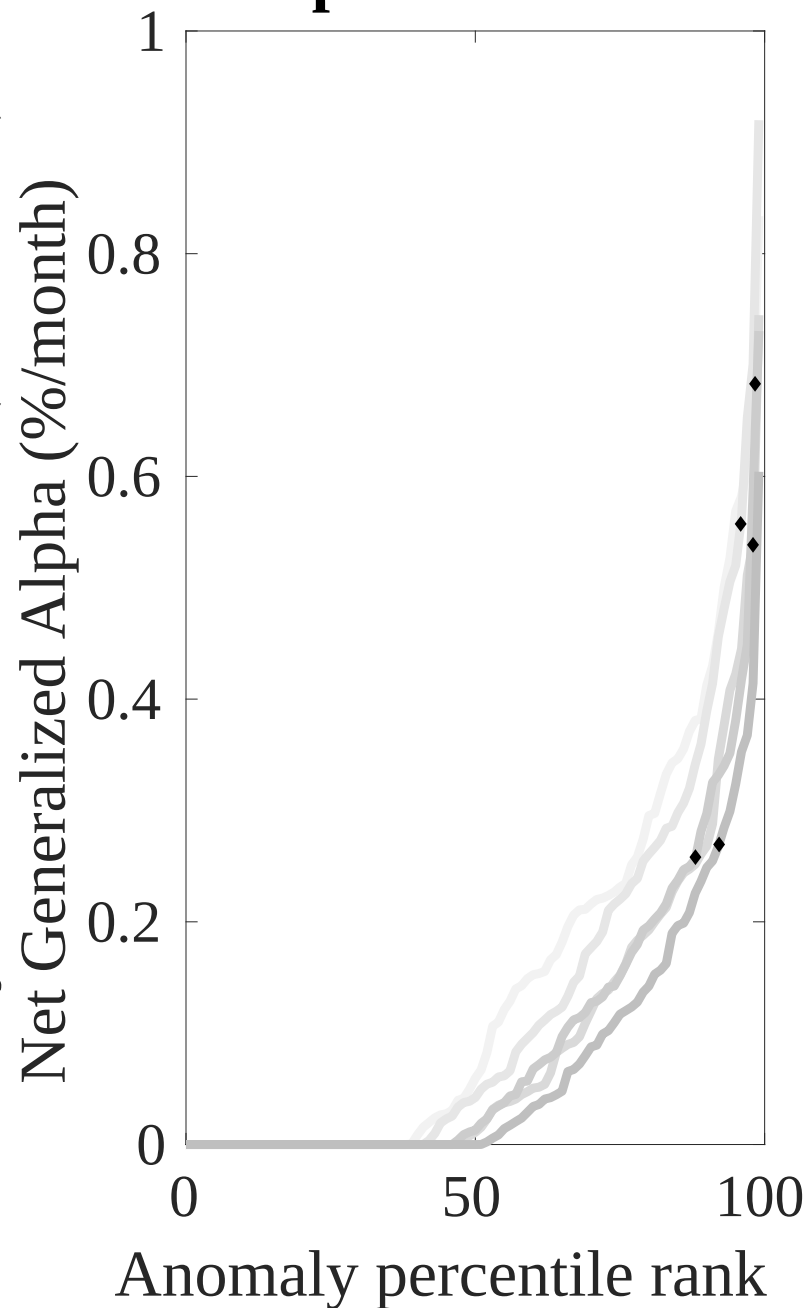
Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles



Novy-Marx and Velikov (ERFS, 2016)

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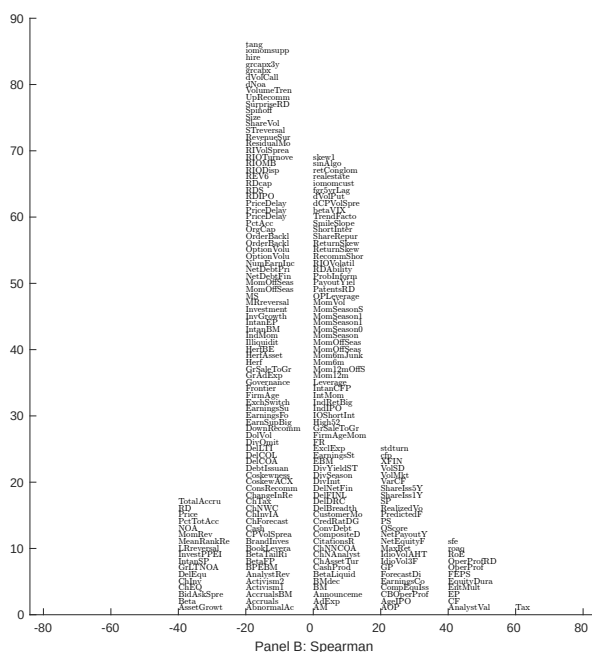
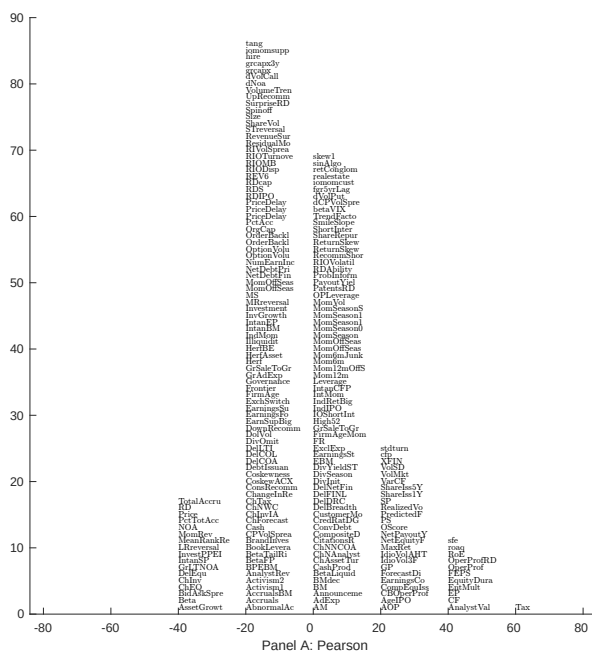
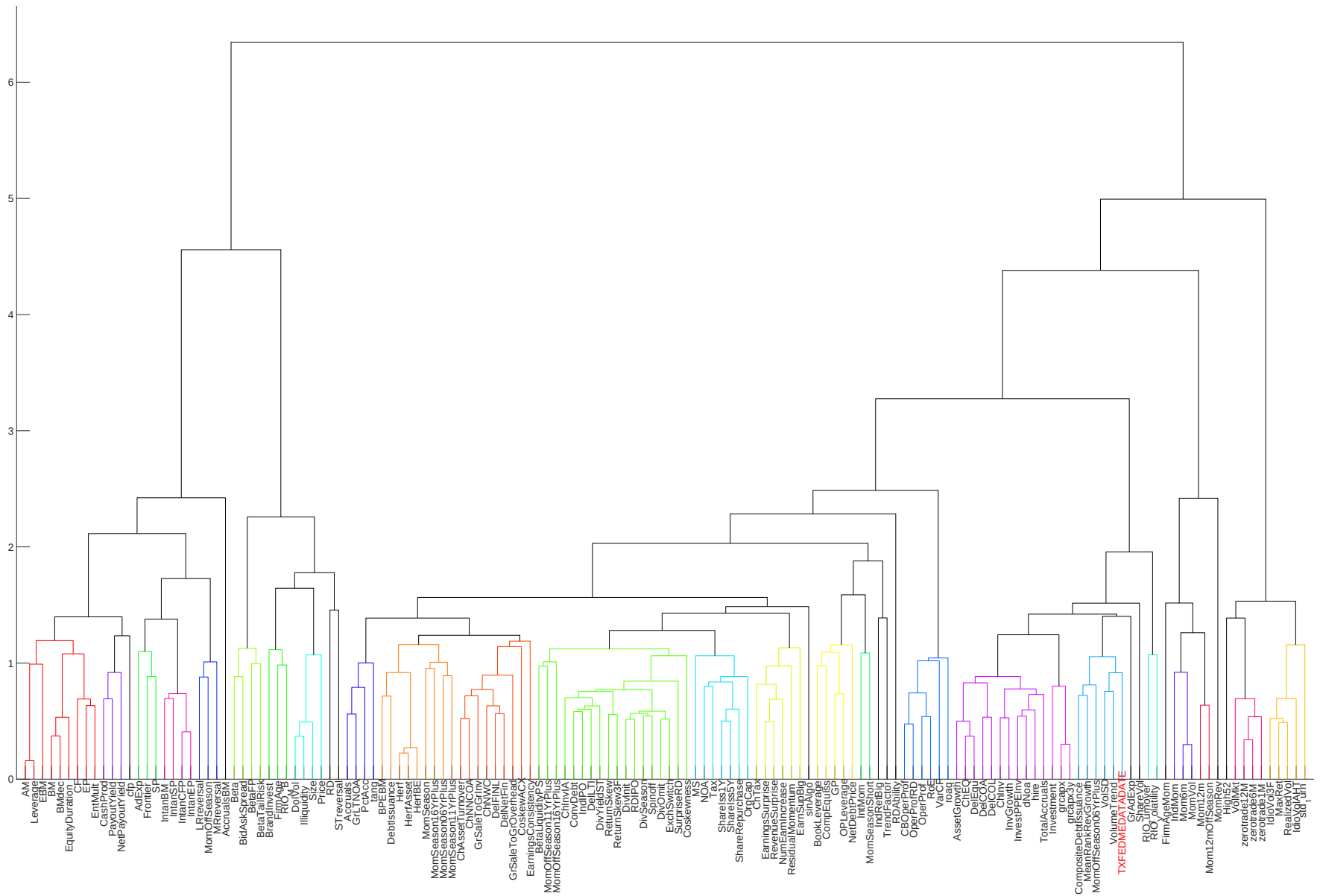


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with TEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



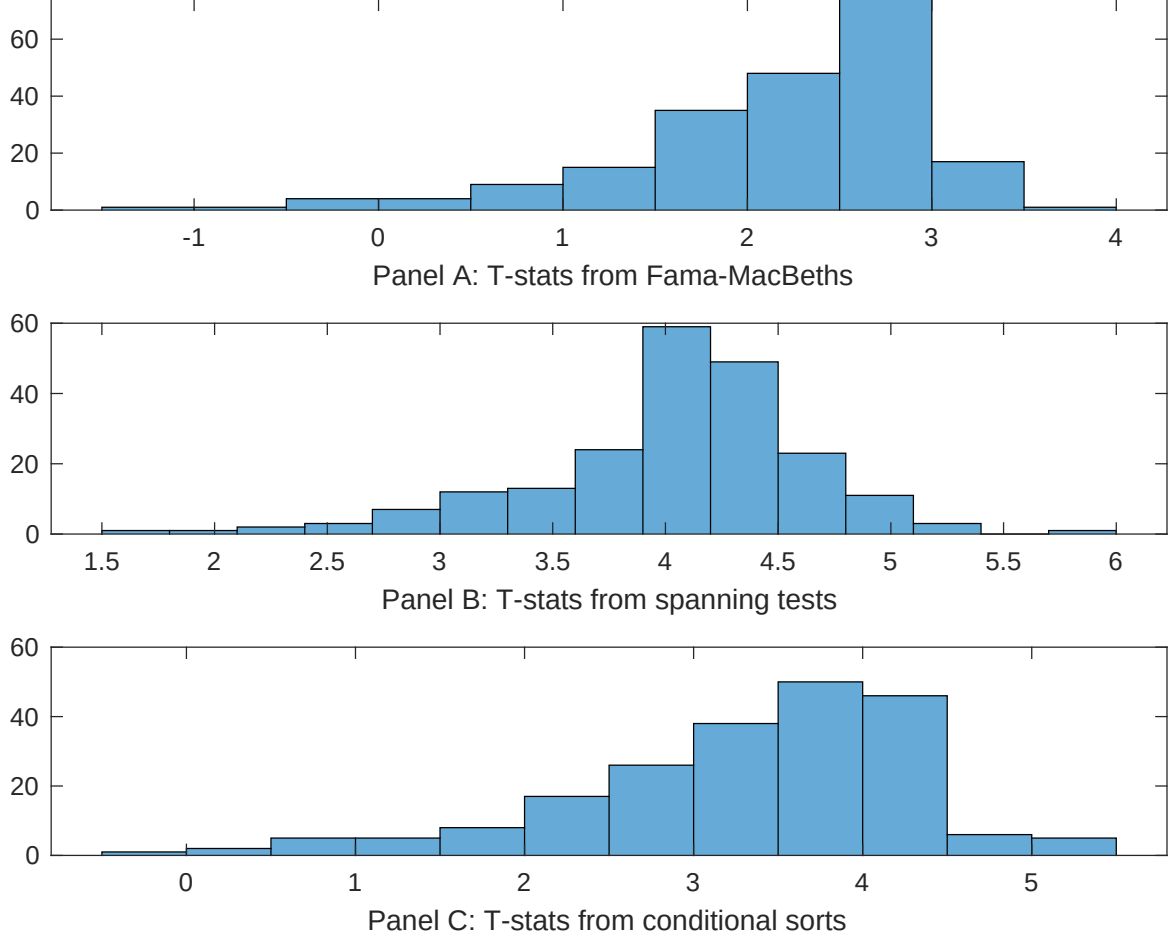


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TEB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97 [3.26]	0.12 [4.85]	0.11 [4.36]	0.14 [7.87]	0.13 [5.04]	0.12 [6.65]	0.11 [5.91]
TEB	0.37 [2.52]	0.26 [1.53]	0.50 [2.50]	0.30 [2.65]	0.14 [0.98]	0.31 [3.16]	0.30 [1.30]
Anomaly 1	0.59 [1.09]						-0.20 [-0.08]
Anomaly 2		0.20 [2.11]					0.52 [1.14]
Anomaly 3			0.31 [2.05]				0.30 [3.09]
Anomaly 4				0.14 [3.44]			0.19 [5.51]
Anomaly 5					0.19 [3.25]		0.44 [1.09]
Anomaly 6						0.93 [1.56]	-0.18 [-2.48]
# months	576	668	576	679	629	679	576
$\bar{R}^2(\%)$	2	1	1	2	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TEB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.28 [2.70]	0.30 [3.06]	0.25 [2.55]	0.25 [2.47]	0.26 [2.38]	0.24 [2.33]	0.26 [2.68]
Anomaly 1	15.71 [3.76]						-3.31 [-0.73]
Anomaly 2		35.65 [9.39]					29.43 [5.41]
Anomaly 3			29.48 [8.52]				27.32 [6.49]
Anomaly 4				19.42 [6.57]			7.23 [1.74]
Anomaly 5					16.36 [3.35]		-15.01 [-2.37]
Anomaly 6						16.68 [4.79]	-1.41 [-0.29]
mkt	-13.44 [-5.10]	-9.78 [-4.15]	-7.72 [-2.99]	-6.17 [-2.25]	-12.91 [-4.86]	-8.99 [-3.29]	-3.37 [-1.24]
smb	17.62 [3.75]	26.31 [7.70]	19.81 [5.08]	34.30 [8.43]	26.47 [6.26]	36.72 [7.48]	22.82 [4.73]
hml	19.21 [3.97]	14.34 [3.05]	26.92 [5.94]	22.50 [4.81]	27.53 [5.69]	24.05 [5.07]	15.99 [3.27]
rmw	69.41 [10.84]	65.67 [13.10]	59.86 [10.96]	71.25 [13.71]	78.96 [13.73]	71.56 [12.58]	51.26 [8.10]
cma	6.89 [0.97]	-17.89 [-2.46]	2.57 [0.38]	1.37 [0.20]	-1.16 [-0.15]	4.39 [0.62]	-12.88 [-1.76]
umd	1.44 [0.56]	3.65 [1.60]	-0.73 [-0.30]	-3.04 [-1.25]	1.43 [0.57]	-1.58 [-0.65]	0.13 [0.05]
# months	577	680	577	680	630	680	577
$\bar{R}^2(\%)$	55	54	59	51	50	50	62

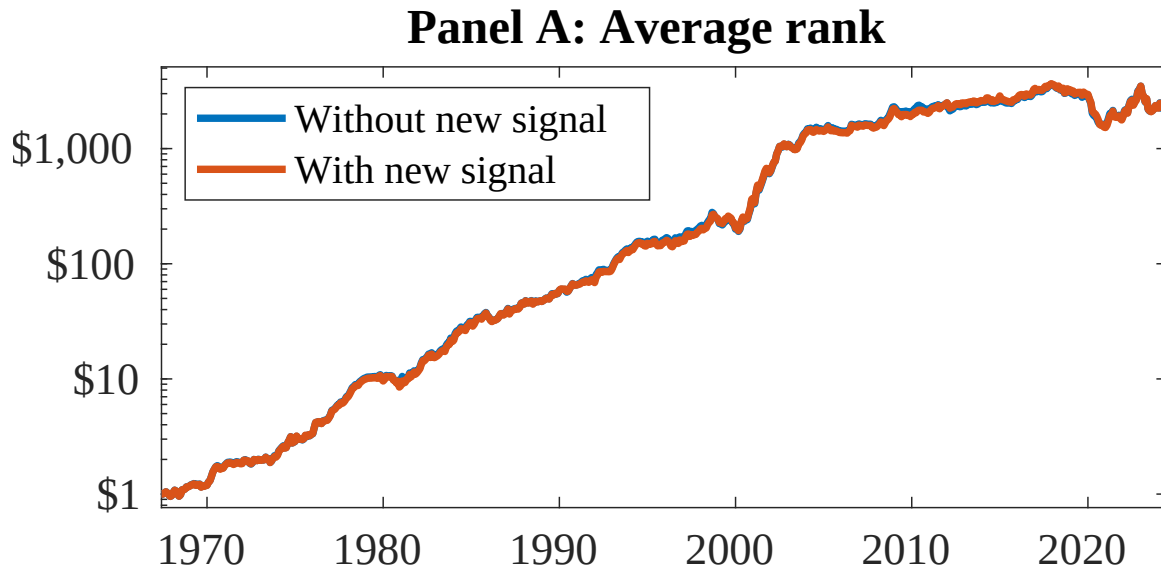


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TEB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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